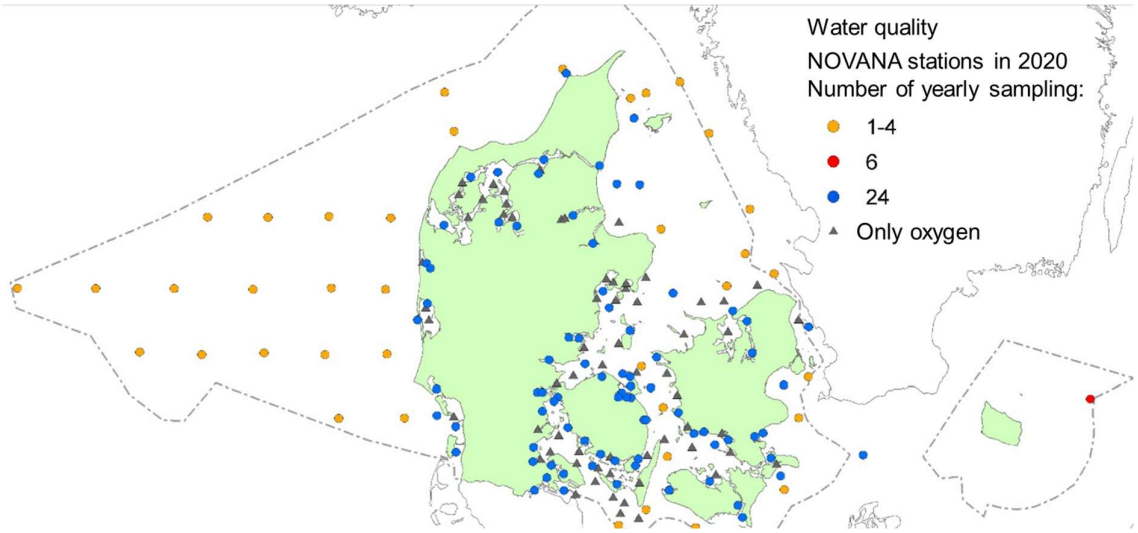
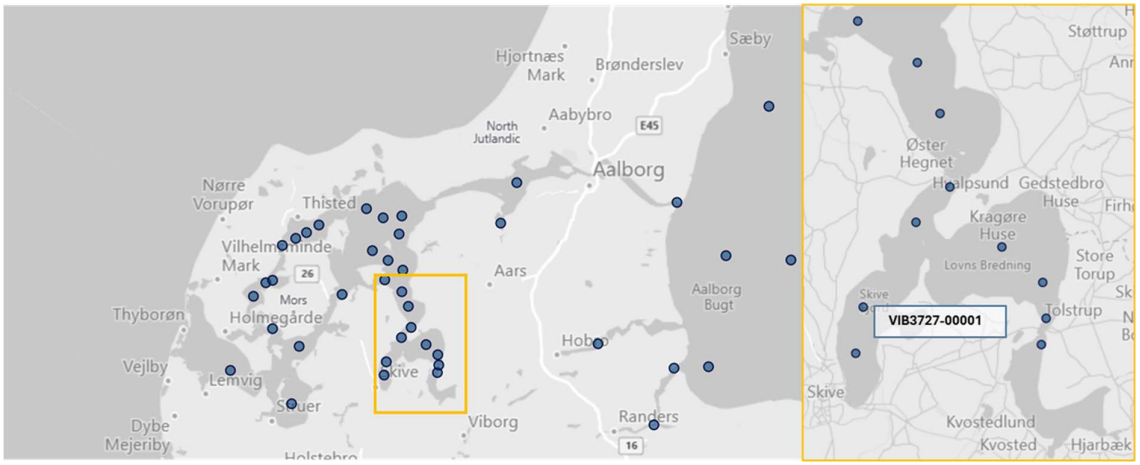
Session 7. Model evaluation

In this exercise, you will visualize water quality data from the Danish national marine monitoring program (NOVANA) and compare the observations against model results from a 3D hydrodynamic-biogeochemical model from DHI to evaluate the model performance.



We will focus our model evaluation on the shallow and eutrophic Skive Fjord in the central part of Limfjorden:



We have selected one monitoring station with complete time series of the water quality dynamics representing Skive Fjord. For this and the following exercises on coastal ecosystem models, we highly encourage you to work together in small groups.

1. **Get familiar with and visualize observations**
   Plot Skive Fjord model output surface and bottom water temperature and salinity as well as dissolved oxygen, total nitrogen and total chlorophyll-a concentrations.

   Two NOVANA monitoring data files are provided:
   SkiveFjord-NOVANA-ctd-2000-2025: Provides information on water temperature, salinity and dissolved oxygen in water column profiles measured with CTD.
   SkiveFjord-NOVANA-wq-2000-2025: Provides information on water quality parameters like nitrate, ammonium, phosphate, total nitrogen and phosphorous (TN and TP), dissolved oxygen and chlorophyll-a concentrations. Most often water samples have been aggregated across parts of the water column, so a mean depth of the sampling depth is provided.
   The datafiles are extracted from the website www.odaforalle.dk and are in Danish. Please ask if you need help with translation.

   Example of NOVANA monitoring water quality data file:

   Station ID | Locality name | Coordinates | Time of sampling Date + Time | Mean depth of sample | Parameter name | Unit | Value

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	DataLeverandørnavn	ObservationsStedNr	ObservationsStedNavn	Lokalitetsnavn	X_UTM32	Y_UTM32	Startdato	Startklok	Prøvetype	GennemsnitsDybde_m	Parameter	Enhed	Prøvefraktion	Resultat	ResultatAtt	Opluknings	Kontrol	Kvalitet
2	Viborg amt	93740004	VIB3726-00001	Risgårde Bredning, Hhvalpsund	510356	6289515	20000106	850	Enkeltprøve	10.6	Ammoniak+ammonium-N	µg/l	Total	71	=	J	C, Faglig k	Godkendt
3	Viborg amt	93740004	VIB3726-00001	Risgårde Bredning, Hhvalpsund	510356	6289515	20000106	850	Enkeltprøve	10.6	Nitrogen,total N	µg/l	Total	890	=	J	C, Faglig k	Godkendt
4	Viborg amt	93740004	VIB3726-00001	Risgårde Bredning, Hhvalpsund	510356	6289515	20000106	850	Enkeltprøve	10.6	Phosphor, total-P	µg/l	Total	43	=	J	C, Faglig k	Godkendt
5	Viborg amt	93740004	VIB3726-00001	Risgårde Bredning, Hhvalpsund	510356	6289515	20000106	850	Enkeltprøve	10.6	Klorofyl a	µg/l	Total	0.55	=	J	C, Faglig k	Godkendt
6	Viborg amt	93740004	VIB3726-00001	Risgårde Bredning, Hhvalpsund	510356	6289515	20000106	850	Enkeltprøve	10.6	Nitrit+nitrat-N	µg/l	Opløst - filtrat fra filtrering	420	=	J	C, Faglig k	Godkendt
7	Viborg amt	93740004	VIB3726-00001	Risgårde Bredning, Hhvalpsund	510356	6289515	20000106	850	Enkeltprøve	10.6	Ortho-phosphat-P	µg/l	Opløst - filtrat fra filtrering	41	=	J	C, Faglig k	Godkendt
8	Viborg amt	93740004	VIB3726-00001	Risgårde Bredning, Hhvalpsund	510356	6289515	20000106	850	Enkeltprøve	1	Oxygen indhold	mg/l	Total	11.6	=	J	C, Faglig k	Godkendt
9	Viborg amt	93740004	VIB3726-00001	Risgårde Bredning, Hhvalpsund	510356	6289515	20000106	850	Enkeltprøve	1	Ammoniak+ammonium-N	µg/l	Total	73	=	J	C, Faglig k	Godkendt
10	Viborg amt	93740004	VIB3726-00001	Risgårde Bredning, Hhvalpsund	510356	6289515	20000106	850	Enkeltprøve	1	Nitrogen,total N	µg/l	Total	1000	=	J	C, Faglig k	Godkendt
11	Viborg amt	93740004	VIB3726-00001	Risgårde Bredning, Hhvalpsund	510356	6289515	20000106	850	Enkeltprøve	1	Phosphor, total-P	µg/l	Total	45	=	J	C, Faglig k	Godkendt
12	Viborg amt	93740004	VIB3726-00001	Risgårde Bredning, Hhvalpsund	510356	6289515	20000106	850	Enkeltprøve	1	Klorofyl a	µg/l	Total	0.62	=	J	C, Faglig k	Godkendt
13	Viborg amt	93740004	VIB3726-00001	Risgårde Bredning, Hhvalpsund	510356	6289515	20000106	850	Enkeltprøve	1	Nitrit+nitrat-N	µg/l	Opløst - filtrat fra filtrering	460	=	J	C, Faglig k	Godkendt
14	Viborg amt	93740004	VIB3726-00001	Risgårde Bredning, Hhvalpsund	510356	6289515	20000106	850	Enkeltprøve	1	Ortho-phosphat-P	µg/l	Opløst - filtrat fra filtrering	38	=	J	C, Faglig k	Godkendt

   Evaluate the water quality dynamics and consider:
   What is the observed mixing regime in Skive Fjord?
   How do the observed oxygen dynamics vary across the season and depth?
   What are the seasonal dynamics of phytoplankton blooms and how eutrophic is Skive Fjord?

2. **Determine fit-for-purpose criteria for a Skive Fjord model**
   The 3D hydrodynamic-biogeochemical for Limfjorden are used to estimate the maximum allowable annual nitrogen and phosphorous input from the catchment to still reach good ecological state according to the Danish implementation of the EU Water Framework Directive.

Formulate fit-for-purpose criteria for a Skive Fjord model. You should include criteria for at least two variables (e.g. salinity and chlorophyll-a concentration). The criteria can be qualitative or quantitative (i.e. use model performance metrics). We encourage you to discuss different criteria within your group.

3. **Visualize model performance**
   We have provided extracted time series for water quality and CTD parameters at the TS_VIB3727 station location from the surface and bottom model output. For the surface and bottom CTD model output we extracted at 0.5 m and 6 m depths.

   Plot observations from the national monitoring program (NOVANA) on top of Skive Fjord surface and bottom model results.

   Tip: For Python users, you can download the package *mikeio* to interact with MIKE data formats (e.g. .dfs0 and .dfsu) directly. Else, we have also provided model results converted to .csv files.

   From just visually inspecting model results against observations, to what degree do you think the model captures the observed water quality dynamics.

4. **Make different validation plots**
   Select 2 or more of the water quality variables you formulated fit-for-purpose criteria and create one or more of the validation plots presented in the lecture.

5. **Optional: Quantify model performance**
   Select 2-4 representative model performance statistics and calculate the model performance statistics for your selected variables.

A. Salinity

Salinity is important because it reflects whether the model reproduces:

- exchange with adjacent waters
- freshwater influence
- vertical stratification and mixing

If the salinity dynamics are wrong, nutrient transport, residence time, oxygen conditions, and phytoplankton dynamics will also be less reliable.

Possible fit-for-purpose criteria for salinity:

Qualitative

- The model should reproduce the observed **seasonal variation** in surface and bottom salinity.
- The model should reproduce periods of **stratification and mixing**.
- The model should capture the general difference between **surface and bottom salinity**.

Quantitative

- Mean bias should be small relative to the observed salinity range.
- Correlation should be sufficiently high to show that the model captures the timing of variability.
- RMSE should be low enough that stratified and mixed periods are distinguishable.

Example criteria:

- correlation $r > 0.7$
- normalized RMSE < 20–25%
- bias within about ±1–2 PSU, depending on observed variability

B. Chlorophyll-a

Chlorophyll-a is a key indicator of eutrophication and ecological state, so it is highly relevant for nutrient management.

Possible fit-for-purpose criteria for chlorophyll-a:

Qualitative

- The model should reproduce the main **seasonal bloom pattern**.
- It should capture the approximate **timing and magnitude** of high-biomass periods.
- It should distinguish between **surface and bottom concentrations**, if observations indicate vertical gradients.

Quantitative

- The model should reproduce seasonal mean concentrations and bloom timing reasonably well.
- Bias and RMSE should be acceptable relative to the large natural variability in chlorophyll.
- Correlation may be lower than for salinity, but should still indicate that the model captures the major dynamics.

Example criteria:

- correlation $r > 0.5$
- normalized RMSE < 30–40%
- bias within ±20–30% of observed mean
- bloom peak timing within a few weeks

C. Oxygen

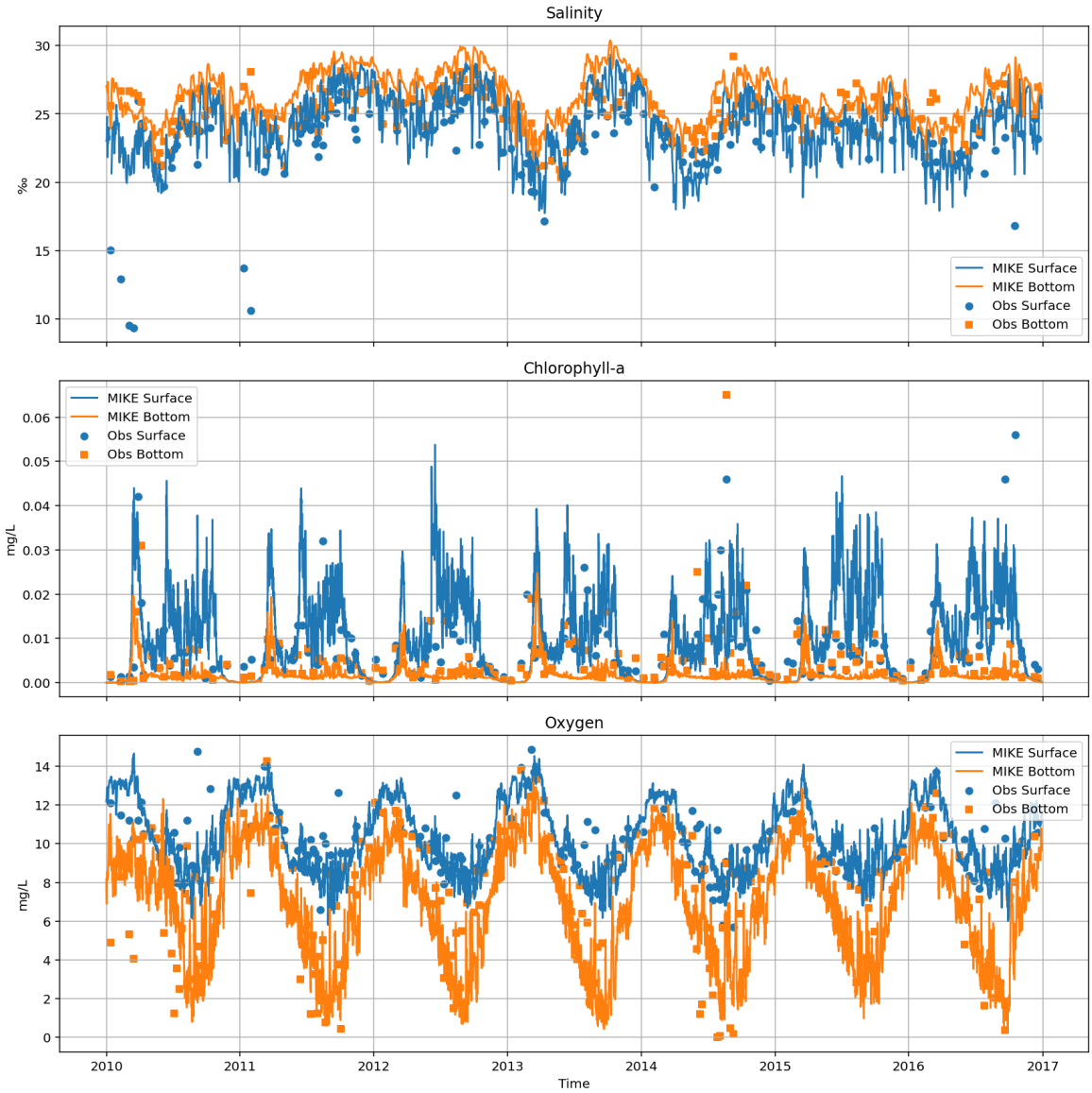
Oxygen is also highly relevant because eutrophication effects often appear through **bottom-water oxygen depletion**.

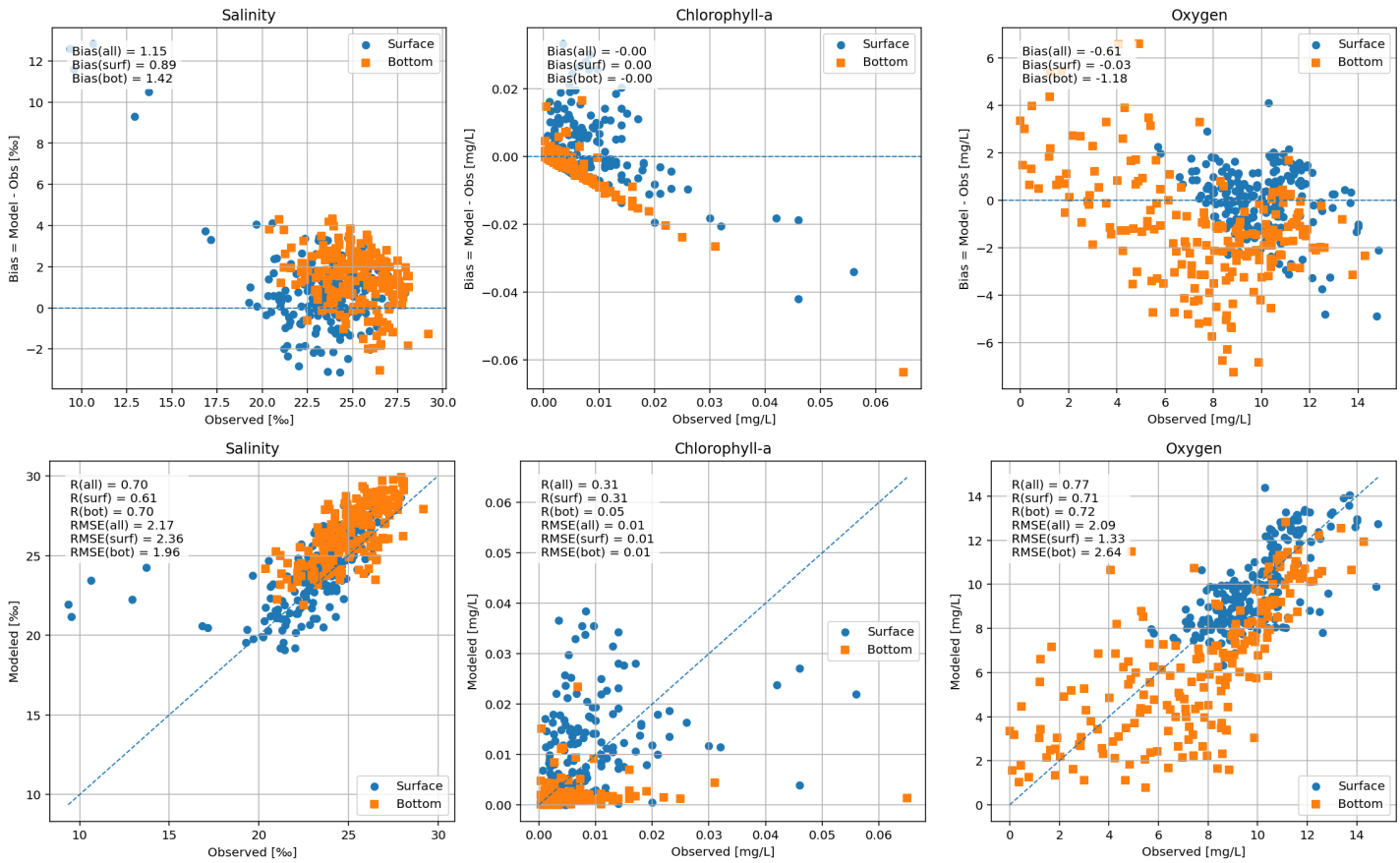
Possible criteria:

- The model should reproduce the seasonal decline in bottom oxygen.
- It should capture the occurrence and duration of low-oxygen periods.
- Bias should be low enough that the model can distinguish between acceptable and problematic oxygen conditions.

Example criteria:

- correlation $r > 0.5$
- bias within ±1–2 mg/L
- correct identification of low-oxygen events





🟦 Salinity

- **R ≈ 0.7 → strong**
- **RMSE ≈ 2‰ → moderate error**
- Bias positive (~ +1‰)

👉 Interpretation:

- Model captures variability well
- Slight systematic overestimation
- Error magnitude is reasonable for estuarine system

🟩 Chlorophyll-a

- **R ≈ 0.3 → weak**
- **RMSE ≈ 0.01 mg/L**

⚠️ BUT:

- Values range ~0–0.06 → RMSE is **not small relative to signal**
- Still clear underestimation of peaks

👉 Interpretation:

- Poor skill despite small absolute RMSE
- Model fails to reproduce bloom dynamics
- RMSE alone misleading → must mention range

🟥 Oxygen

- **R ≈ 0.77 → strong**
- **RMSE ≈ 2.1 mg/L → noticeable error**
- Bottom RMSE much larger (2.64 vs 1.33)

👉 Interpretation:

- Good dynamics captured
- Larger errors at bottom → sediment / respiration processes not well represented

🧠 Key insights (what really matters)

1. Physics vs Biology

- **Salinity → good (physics works)**
- **Oxygen → decent (coupled processes OK-ish)**
- **Chl-a → poor (biology weak)**

👉 This is a *very standard and good conclusion*

2. Depth dependence (important!)

- Bottom always worse:
  - higher RMSE
  - stronger bias

👉 You should explicitly say:

> Model performance decreases toward the bottom layer

3. RMSE vs R insight (nice for report)

- Chlorophyll:
  - low RMSE but low R → model gets magnitude roughly but not variability
- Oxygen:
  - high R but higher RMSE → good pattern, imperfect magnitude

👉 This is **exactly the kind of discussion they want**